

Class Number Formula

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Prerequisites

Let K be the number field and the corresponding number ring $R = \mathbb{A} \cap K$. For each positive real t , we denote the number of ideals with $||I|| \leq t$ as $i(t)$ and in the ideal class C with $||I|| \leq t$ as $i_C(t)$. Thus $i(t) = \sum_C i_C(t)$ which is finite. Let $n = [K : \mathbb{Q}]$.

Theorem

There exists a real constant κ , only depending on R , such that

$$i_C(t) = \kappa t + \epsilon_C(t)$$

where the error term $\epsilon_C(t)$ is $O(t^{1-1/n})$.

Corollary

$i(t) = h\kappa t + \epsilon(t)$ where h is the no. of ideal classes in R and $\epsilon(t)$ is $O(t^{1-1/n})$

Sketch of Proof

The theorem gives us that $\kappa = \lim_{t \rightarrow \infty} \frac{i_C(t)}{t}$. Calculating $i_C(t)$ is equivalent to counting the number of elements in an ideal, and so there is a one-one correspondence:

$$\{I \in C; ||I|| \leq t\} \longleftrightarrow \{\alpha \in J; ||(\alpha)|| \leq t||J||\}$$

We construct a subset of R such that no two elements are associates and every non-zero element in R has a unit multiple. Such a set is the coset representatives of the group of units U of $R \setminus \{0\}$, and we count the elements of J in that set. By Dirichlet's Unit Theorem, $U = W \times V$ where W is the cyclic group of roots of unity in K and U is the free abelian subgroup of rank $r + s - 1$. Fixing V , we look at the map

$$V \subset U \subset R \setminus \{0\} \hookrightarrow \Gamma_R \setminus \{0\} \rightarrow \mathbb{R}^{r+s}$$

Here U maps onto $\Gamma_U \subseteq H \subseteq \mathbb{R}^{r+s}$ and H is hyperplane of $y_1 + \dots + y_{r+s} = 0$. Kernel of $U \rightarrow \Gamma_U$ is W and the restriction $V \rightarrow \Gamma_U$ is isomorphism.

Replace $\mathbb{R}^n \setminus \{0\}$ by $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$. By improving the maps we get

$$V \subset U \subset \mathbb{R} \setminus \{0\} \hookrightarrow (\mathbb{R}^*)^r \times (\mathbb{C}^*)^s \xrightarrow{\log} \mathbb{R}^{r+s}$$

The first map takes $\alpha \in \mathbb{R} \setminus \{0\} \mapsto (\sigma_1(\alpha), \dots, \sigma_r(\alpha), \gamma_1(\alpha), \dots, \gamma_s(\alpha))$
 And $\log(x_1, \dots, x_r, z_1, \dots, z_s) = (\log|x_1|, \dots, 2\log|z_1|, \dots)$.

Coset Representatives for V can be obtained from a set of coset representatives for V' in group $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$ is called fundamental domain for V' say D .

Define $N(x_1, \dots, x_r, z_1, \dots, z_s) = x_1 \cdots x_r |z_1|^2 \cdots |z_s|^2$ in $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$.

This $|N(x)| = ||(\alpha)||$. We want:

- To find D for V' in $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$
- Counting elements $x \in \Gamma_J \cap D$ with $|N(x)| \leq t||J||$

So, no. of $x = |W| \cdot i_C(t)$

Lemma 1

Let $f : G \rightarrow G'$ be a homomorphism of abelian groups and let S be a subgroup of G carried isomorphically onto $S' \subseteq G'$. Suppose D' is a set of coset representatives for S' in G' . Then its total inverse image $D = f^{-1}(D')$ is a set of coset representatives for S in G .

Applying lemma 1 to log function, V' maps onto Γ_U . So need a set D' of coset representatives for Γ_U in $\mathbb{R}^{r+s}$. Its total inverse image is D .

In general fix fundamental parallelepiped F for Γ_U in H and fix vector v passing through O and orthogonal to H say $v = (1, \dots, 1, 2, \dots, 2)$. Then

$$D = \{x \in (\mathbb{R}^*)^r \times (\mathbb{C}^*)^s : \log x \in F \oplus \mathbb{R}v (= D')\}$$

Here $D = aD$ for all $a \in \mathbb{R}, a \neq 0$. Define $D_a = \{x \in D : |N(x)| \leq a\}$.

Using this we get $D_a = \sqrt[n]{a} D_1$. So we need to count points in

$$\Gamma_J \cap \sqrt[n]{t} \|J\| D_1$$

Theorem

Let Γ be an n -dimensional lattice in $\mathbb{R}^n$ and let B be a bounded subset of $\mathbb{R}^n$ with a "sufficiently nice" boundary. Then, for $a \in \mathbb{R}$,

$$|\Gamma \cap aB| = \frac{\text{Vol}(B)}{\text{Vol}(\Gamma)} a^n + \gamma(a),$$

where $\gamma(a) = O(a^{n-1})$.

Here "sufficiently nice" boundary means $(n-1)$ Lipschitz parametrizable i.e boundary is covered by images of finitely many Lipschitz functions $f : [0, 1]^{n-1} \rightarrow \mathbb{R}^n$. Applying the theorem to Γ_J as D_1 has such boundary and using $\text{Vol}(\Gamma_J) = \text{Vol}(\Gamma_R) \|J\|$,

$$|\Gamma_J \cap \sqrt[n]{t \|J\|} D_1| = \frac{\text{Vol}(D_1)}{\text{Vol}(\Gamma_R)} t + O(t^{1-1/n})$$

Formula for κ

Hence $i_C(t) = \kappa t + O(t^{1-1/n})$ where

$$\kappa = \frac{\text{vol}(D_1)}{|W| \cdot \text{vol}(\Gamma_R)} = \frac{2^s \text{vol}(D_1)}{|W| \sqrt{|\text{disc}(R)|}} \quad \square$$

Now let D_1^+ consists of all points of D_1 s.t $x_1, \dots, x_r \geq 0$. Then $\text{vol}(D_1) = 2^r \text{vol}(D_1^+)$. Let the fundamental parallelepiped F has form $\sum_{k=1}^{r+s-1} t_k v_k : 0 \leq t_k \leq 1$ for some $\mathbb{Z}$ basis $\{v_1, \dots, v_{r+s-1}\}$ for Γ_U . For each k , set $v_k = (v_k^{(1)}, \dots, v_k^{(r+s)})$. Now a point $(x_1, \dots, x_r, z_1, \dots, z_s)$ in D_1^+ is characterized by:

$$\begin{aligned} \log(x_1) &= \sum_{k=1}^{r+s-1} t_k v_k^{(1)} + u \dots \log(x_r) = \sum_{k=1}^{r+s-1} t_k v_k^{(r)} + u \\ 2\log|z_1| &= \sum_{k=1}^{r+s-1} t_k v_k^{(r+1)} + 2u \dots 2\log|z_s| = \sum_{k=1}^{r+s-1} t_k v_k^{(r+s)} + 2u \end{aligned}$$

Now any element $(x_1, \dots, x_r, z_1, \dots, z_s)$ of D_1^+ has $|N(x)| \leq 1$ and if the log of first $r + s - 1$ belongs to F then

$$\begin{aligned} \log(x_1), \dots, \log(z_{s-1}) \geq 0 &\implies |x_1|, \dots, |x_r|, |z_1|^2, \dots, |z_{s-1}|^2 \geq 1 \\ &\implies |z_s| \leq 1 \implies u \leq 0 \end{aligned}$$

Set $t_{r+s} = e^u$ and introduce polar coordinates (ρ_j, θ_j) for each z_j we get D_1^+ as $(x_1, \dots, x_r, \rho_1 e^{i\theta_1}, \dots, \rho_s e^{i\theta_s})$ satisfying following equations

$$x_j = t_{r+s} e^{\sum_{k=1}^{r+s-1} t_k v_k^j}, \quad \rho_j = t_{r+s} e^{\frac{1}{2} \sum_{k=1}^{r+s-1} t_k v_k^{r+j}}, \quad \theta_j = 2\pi t_{r+s+j}$$

With $t_{r+s} \in (0, 1]$ and all other $t_k \in [0, 1)$

Now to get formula for κ

$$\begin{aligned} \text{vol}(D_1^+) &= \int_{D_1^+} \rho_1 \cdots \rho_s dx_1 \cdots dx_r d\rho_1 \cdots d\rho_s d\theta_1 \cdots d\theta_s \\ &= \int_{[0,1]^n} \rho_1 \cdots \rho_s |J(t_1, \dots, t_n)| dt_1 \dots dt_n \end{aligned}$$

where $J = \frac{\pi^s x_1 \cdots x_r \rho_1 \cdots \rho_s}{t_{r+s}} \det(M)$ and $M = \begin{pmatrix} v_1^{(1)} & \cdot & \cdot & v_k^{(r+s)} \\ \cdot & \cdot & \cdot & \cdot \\ v_{r+s-1}^{(1)} & \cdot & \cdot & v_{r+s-1}^{(r+s)} \\ 1 & \cdot & \cdot & 2 \end{pmatrix}$

After calculating we get $\text{vol}(D_1^+) = \frac{1}{n} \pi^s |\det(M)|$.

We define $\text{Reg}(R) = \frac{1}{n} \det(M)$. So we have proved $\kappa = \frac{2^{r+s} \pi^s \text{Reg}(R)}{|W| \sqrt{|\text{disc}(R)|}}$

For imaginary quadratic case there is no free part, so by default $\text{Reg}(R) = 1$ and $s = 1, r = 0$. So $\kappa = \frac{\pi}{\sqrt{|\text{disc}(R)|}}$ for $d \neq (-1), (-3)$ and $\frac{\pi}{2\sqrt{|\text{disc}(R)|}}, \frac{\pi}{3\sqrt{|\text{disc}(R)|}}$ for $d = (-1), (-3)$ resp.

For real quadratic field there is a free part of rank 1 say u . And $r = 2, s = 0$. Now

$$u \mapsto (\sigma_1(u), \sigma_2(u)) \mapsto (\log|\sigma_1(u)|, \log|\sigma_2(u)|)$$

And $|\sigma_1(u)\sigma_2(u)| = 1$ with $\sigma_1(u) = u \implies \log u(1, -1)$ forms a lattice in Γ_U . So calculating we get $\text{Reg}(R) = \log(u)$ and $\kappa = \frac{2\log(u)}{\sqrt{|\text{disc}(R)|}}$

Dedekind Zeta Function

Consider Dirichlet series $\sum_{n=1}^{\infty} \frac{a_n}{n^s}$ where $a_n \in \mathbb{C}$ and $s = x + iy$.

Weierstrass's Theorem

If $f_n(z)$ is analytic seqⁿ in Ω_n and $\{f_n(z)\} \rightarrow f(z)$ in Ω uniformly on every compact set of Ω . Then $f(z)$ is analytic in Ω . Moreover $f'_n(z) \rightarrow f'(z)$ on every cpt sets of Ω

Proposition

Suppose $\sum_{n \leq t} a_n = O(t^r)$ for some real $r \geq 0$. Then the dirichlet series formed by $\{a_n\}$ converges for all s with $x > r$, and is an analytic function of s on that half plane.

Applying the above proposition to Reimann zeta function

$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, it converges and analytic on half plane $x > 1$. Now we define Dedekind zeta function ζ_K of a number field K for $x > 1$ by

$$\zeta_K(s) = \sum_{n=1}^{\infty} \frac{j_n}{n^s}$$

where $j_n = \text{no. of } I \text{ of } R = \mathbb{A} \cap K \text{ with } ||I|| = n.$

We know $\sum_{n \leq t} j_n = i(t) = O(t)$ hence the $Prop^n$ says ζ_K converges and is analytic on half plane $x > 1$. We know $\zeta(s)$ can be extended to a analytic function for $x > 0$ except at $s = 1$ where there is simple pole. For extending $\zeta_K(s)$ we have

$$\zeta_K(s) = \sum_{n=1}^{\infty} \frac{j_n - h\kappa}{n^s} + h\kappa\zeta(s)$$

for $x > 1$, where h is ideal class number in R . Applying the $Prop^n$

$$\sum_{n \leq t} j_n - h\kappa = i(t) - h\kappa t = O(t^{1 - \frac{1}{[K:\mathbb{Q}]}})$$

Hence ζ_K is analytic for $x > 1 - \frac{1}{[K:\mathbb{Q}]}$ except for $s=1$ where it has simple pole. As ζ_K absolutely converges for $x > 1$ we can rearrange terms to get

$$\zeta_K(s) = \sum_I \frac{1}{||I||^s} \quad \text{for } x > 1$$

Lemma

Let $a_1, a_2, \dots, a_n \in \mathbb{C}$ be a sequence s.t $1+a_n \neq 0$ then $\prod_{n=1}^{\infty} (1 + a_n)$ converges absolutely iff $\sum_{n=1}^{\infty} |a_n|$ converges absolutely.

Theorem

Let $a_1, \dots, a_n \in \mathbb{C}$, $|a_i| < 1 \forall i$ and $\sum_{i=1}^{\infty} |a_i| < \infty$. Then

$$\prod_{i=1}^{\infty} (1 - a_i)^{-1} = 1 + \sum_{j=1}^{\infty} \sum_{(r_1, \dots, r_j)} a_1^{r_1} \dots a_j^{r_j}$$

The sum and product are both absolutely convergent, hence independent of the order of a_i .

Now every I has a unique factorization in R in terms of P prime ideals. Hence

$$\sum |||P||^{-s} = \sum ||P||^{-x} < \sum ||I||^{-x} < \infty$$

As $\zeta_K(x)$ converges for $x > 1$. Hence applying the theorem we get

$$\zeta_K(s) = \prod_P \left(1 - \frac{1}{||P||^s}\right)^{-1} \quad \text{for } x > 1$$

Now if we look at $\frac{\zeta_K(s)}{\zeta(s)}$ then its first part is analytic at $s = 1$ so

$$\rho = \lim_{s \rightarrow 1} \frac{\zeta_K(s)}{\zeta(s)} = h\kappa.$$

We will calculate h for abelian extension K , so K is galois extension. For each prime $p \in \mathbb{Z}$, we denote r_p as the number of primes in R lying over p , and let the inertial degree $f(P|p)$ be denoted as f_p . Then,

$$\zeta_K(s) = \prod_p \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p}, \quad \text{for } x > 1$$

Class number formula

We recall the definition of L -functions:

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}$$

where χ is the character modulo m , where $K \subset \mathbb{Q}(e^{\frac{2\pi i}{m}})$ (by Kronecker Weber Theorem) which converges for $x > 0$ for non trivial character and for trivial character it converges for $x > 1$. So,

$L(s, 1) = \zeta(s) \prod_{p|m} \left(1 - \frac{1}{p^s}\right)$. Let G be the Galois group of $K / \mathbb{Q}$. Now, $G \subset \mathbb{Z}_m^* \implies \hat{G} \subset \hat{\mathbb{Z}}_m^*$. Fixing prime $p \nmid m$, we also have formula $\prod_{\chi \in \hat{G}} \left(1 - \frac{\chi(p)}{p^s}\right) = \left(1 - \frac{1}{p^{f_p s}}\right)^{r_p}$, for $x > 1$, from which it follows that

$$\frac{\zeta_K(s)}{\zeta(s)} = \prod_{p|m} \left(1 - \frac{1}{p^s}\right) \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p} \prod_{1 \neq \chi \in \hat{G}} L(s, \chi)$$

As, for $\chi \neq 1$, $L(s, \chi)$ is analytic for $\sigma > 0$, we have the required formula for ρ just by putting $s = 1$

$$\rho = \prod_{p|m} \left(1 - \frac{1}{p^s}\right) \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p} \prod_{1 \neq \chi \in \hat{G}} L(1, \chi)$$

Theorem

Let χ be a non-trivial character modulo m . Then

$$L(1, \chi) = -\frac{1}{m} \sum_{k=1}^{m-1} \tau_k(\chi) \log(1 - \omega^{-k})$$

where $\tau_k(\chi) = \sum_{a \in \mathbb{Z}_m^*} \chi(a) \omega^{ak}$ and $\log(1 - z) = -\sum_{n=1}^{\infty} z^n/n$ for all $z < 1$ and also for those z on unit circle for which series converges.

Proof For s in half plane $x > 1$ we have,

$$\begin{aligned}
 L(s, \chi) &= \sum_{a \in \mathbb{Z}_m^*} \chi(a) \sum_{n \equiv a \pmod{m}, n \geq 1} \frac{1}{n^s} \\
 &= \sum_{a \in \mathbb{Z}_m^*} \chi(a) \sum_{n=1}^{\infty} \frac{\frac{1}{m} \sum_{k=0}^{m-1} \omega^{(a-n)k}}{n^s} \\
 &= \frac{1}{m} \sum_{k=0}^{m-1} \tau_k(\chi) \sum_{n=1}^{\infty} \frac{\omega^{-nk}}{n^s}
 \end{aligned}$$

by absolute convergence exchange of sum possible. Also,

$\tau_0(\chi) = \sum_{a \in \mathbb{Z}_m^*} \chi(a) = 0$, so $1 \leq k \leq m-1$. And for such k , $\sum_{n=1}^{\infty} \frac{\omega^{-nk}}{n^s}$ converges to analytic function on $x > 0$ and so cont. at $s = 1$ and equal to $-\log(1 - \omega^{-k})$. So we get the theorem just by approaching 1 from right. .

Contd.

We can further simplify the R.H.S. in the previous theorem using the values of τ_k , as given by the following lemma.

Lemma

Let χ be primitive character modulo m . Then

$$\tau_k(\chi) = \begin{cases} \overline{\chi}(k)\tau(k), & \text{if } (k, m) = 1 \\ 0, & \text{if } (k, m) > 1 \end{cases}$$

where $\tau(k) = \tau_1(k)$.

Substituting the values of τ_k in the expression for $L(1, \chi)$, and calculating for odd and even characters separately, we get

$$|L(1, \chi)| = \begin{cases} \frac{2}{\sqrt{m}} \left| \sum_{k \in \mathbb{Z}_m^*; k < m/2} \chi(k) \log \sin \frac{k\pi}{m} \right|, & \chi \text{ is even} \\ \frac{\pi}{|2 - \chi(2)|\sqrt{m}} \left| \sum_{k \in \mathbb{Z}_m^*} \chi(k) \right|, & \chi \text{ is odd} \end{cases}$$

This gives us the complete formula.

Quadratic Case

We will calculate the class number formula for the field $K = \mathbb{Q}[\sqrt{d}]$, where d is a square-free integer. We assume that $m = |\text{disc}(R)|$. So,

$$m = \begin{cases} |d|, & \text{if } d \equiv 1 \pmod{4} \\ |4d|, & \text{if } d \not\equiv 1 \pmod{4} \end{cases}$$

Also we can prove that $K \subset \mathbb{Q}(e^{2\pi i/m})$. Also, p is ramified in R if and only if $p \mid m$.

We note that in this case, $r_p = 1 = f_p$. Hence, using the formula of ρ ,

$$\rho = \prod_{p|m} \left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{p^{f_p}}\right)^{-r_p} \prod_{1 \neq \chi \in \hat{G}} L(1, \chi) = L(1, \chi)$$

where χ is the unique non-trivial character modulo m corresponding to the character of the Galois group of K over $\mathbb{Q}$.

Let us define the character as following: for odd prime $p \nmid m$,

$$\chi(p) = \left(\frac{d}{p}\right)$$

Quadratic Case

If $2 \nmid m$, which means that $d \equiv 1 \pmod{4}$,

$$\chi(2) = \begin{cases} 1, & \text{if } d \equiv 1 \pmod{8} \\ -1, & \text{if } d \equiv 5 \pmod{8} \end{cases}$$

The above two definitions taken together give us the multiplicative homomorphism $\chi : \mathbb{Z}_m^* \rightarrow \mathbb{C}$, defined by $\chi(n) = \left(\frac{d}{n}\right)$, for odd integers n , where $(.)$ denotes the Jacobi symbol.

Theorem

For Quadratic fields, we have if p is odd prime

If $p \mid d$, then

$$pR = (p, \sqrt{d})^2$$

If $p \nmid d$, then

$$pR = \begin{cases} (p, n + \sqrt{d})(p, n - \sqrt{d}) & \text{if } d \equiv n^2 \pmod{p} \\ \text{prime} & \text{if } d \not\equiv n^2 \pmod{p} \end{cases}$$

Applying the above theorem for $p \nmid d$ i.e $p \nmid m$ (as d and m has same prime factors) case, we get $r_p = 2, f_p = 1$ for $(\frac{d}{p}) = 1$ and $r_p = 1, f_p = 2$ for $(\frac{d}{p}) = -1$. Hence,

$$\begin{aligned} \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p} &= \left(1 - \frac{1}{p^s}\right)^{-1} \left(1 - \frac{(\frac{d}{p})}{p^s}\right)^{-1} \\ \implies \zeta_K(s) &= \zeta(s) \prod_p \left(1 - \frac{(\frac{d}{p})}{p^s}\right)^{-1} \\ \implies \frac{\zeta_K(s)}{\zeta(s)} &= L(s, \chi) \end{aligned}$$

Similar for $p = 2$ case we can verify the definition. Also χ here comes out to be primitive character mod m hence we can apply the previous thm in calculation. Moreover, χ is even if $d > 0$ and odd if $d < 0$.

Examples

So final class formula for quadratic case can be given by $h = \frac{\rho}{\kappa}$,

$$h = \begin{cases} \frac{1}{\log u} \left| \sum_{k \in \mathbb{Z}_m^*; k < m/2} \chi(k) \log \sin \frac{k\pi}{m} \right|, & d > 0 \\ \frac{1}{|2 - \chi(2)|\sqrt{m}} \left| \sum_{k \in \mathbb{Z}_m^*} \chi(k) \right|, & d < 0, d \neq -1, -3 \end{cases}$$

For $d = -1, -3$ case we get additional factor 2,3 resp. with the given formula.

Examples

1) $d = -2$: Here $m = 8$ as $d \not\equiv 1 \pmod{4}$. $\chi(3) = \left(\frac{-2}{3}\right) = \left(\frac{1}{3}\right)$. Hence,

$$h = \frac{1}{2} |\chi(1) + \chi(3)| = 1$$

2) $d = -5$: Here

$m = 20, \chi(3) = \left(\frac{-5}{3}\right) = \left(\frac{1}{3}\right) = 1, \chi(7) = \left(\frac{-5}{7}\right) = \left(\frac{2}{7}\right) = 1, \chi(9) = \left(\frac{-5}{9}\right) = 1$

$$h = \frac{1}{2} |\chi(1) + \chi(3) + \chi(7) + \chi(9)| = 2$$

Examples

3) $d = 2$: Here $m = 8, \chi(3) = -1, u = (1 + \sqrt{2})$

$$h = \frac{1}{\log(1+\sqrt{2})} |\chi(1) \log \sin(\frac{\pi}{8}) + \chi(3) \log \sin(\frac{3\pi}{8})| = 1$$

Since,

$$(1 + \sqrt{2}) = \frac{\sin \frac{3\pi}{8}}{\sin \frac{\pi}{8}}$$

4) $d = 3$: Here $m = 12, u = (2 + \sqrt{3}), \chi(5) = (\frac{3}{5}) = -1$

$$h = \frac{1}{\log(2+\sqrt{3})} |\chi(1) \log \sin(\frac{\pi}{12}) + \chi(5) \log \sin(\frac{5\pi}{12})| = 1$$

- Number Fields by Daniel Marcus
- Complex Analysis by Lars Ahlfors

Thank You For Your Attention!